

Crime Prediction Fairness Project — What We Did, Step by Step

A plain-English record of every stage, the result, and what changed because of it. Prepared for review. All model results below come from the final, reproducibility-verified code unless marked otherwise.

Part 1 — The full journey

Step	What we did	What we found	What it meant	Outcome
1	Rebuilt the FedCrime paper Wrote the whole model from the paper: ZINB loss, TCN time model, federated training across 6 police stations.	Matched the paper on real Los Angeles data. Accuracy fell from 72.2 to 48.4 as the data got sparser, same as the authors reported.	The starting point was trustworthy. We could now build on it.	Baseline established
2	Added 9 comparison methods FedAvg, FedProx, SCAFFOLD, MOON, FBLG, FedAvgM, TrimAvg, Multi-Krum, Bulyan.	All ran and produced comparable numbers.	We could compare our work against what the field already uses.	Baselines built
3	Added a GNN (professor's suggestion) Let neighbouring areas share information.	On a small made-up test set it looked like a big win (+17 points). On real city data the gain vanished.	First lesson: toy-data results do not transfer. We stopped trusting small tests.	Warning sign
4	Found the fairness problem Split areas into Head (busiest), Mid, Tail (quietest) and measured each.	Huge gap: Head scored about 69, Tail about 28. A 41-point difference.	Looked like the model was badly failing poor neighbourhoods. This became the research question.	Problem identified
5	Tried three standard fixes (a) give quiet areas more weight, (b) Group-DRO, (c) tune the decision threshold per area.	(a) made it worse. (b) stuck at ~41 for every setting tried. (c) improved to ~41 then stopped.	Three reasonable fixes all failed. That suggested the problem was not what we assumed.	All three failed
6	The key idea: the gap may be fake Worked out the maths of the F1 score in rare-event data.	A model with zero skill scores $2p/(1+p)$ on F1, where p is how often crime happens. Busy areas therefore score high automatically.	The Head-Tail gap can appear even when the model treats both areas identically.	Diagnosis
7	Proved it in simulation	Gave two groups identical skill by construction. F1 gap: 38.7 points. AUC gap: 0.19 points.	Confirmed: F1 invents a gap that is not there. AUC does not.	Theory verified
8	Checked 4 cities LA, Chicago, New York, San Francisco.	Fake gap: LA 42.8, Chicago 45.9, NYC 19.9, SF 48.8. Correlation with sparsity $r = +0.835$.	The sparser the city's data, the bigger the fake gap. Consistent everywhere.	Holds across cities

Step	What we did	What we found	What it meant	Outcome
9	Audited 7 published papers Took other researchers' reported numbers and checked them against crime volume.	Their scores tracked crime volume at $r = 0.899$. FedCrime's own tables: $r = 0.998$.	Published fairness gaps in this field are largely explained by base rates, not model quality.	Affects the literature
10	Checked a non-crime dataset Chicago 311 service requests, same areas.	Same pattern appeared.	Not a crime-data quirk. It is a property of the metric.	Generalises
11	Asked: does it change decisions?	Choosing the best model: unaffected (0 of 5 changed). Choosing which area needs help: changed for 6 of 6 models.	The flaw does not mislead you about which method to use. It misleads you about <i>where</i> the model is failing.	Real consequence
12	Built a tool (fairaudit)	One command runs the whole check on any dataset.	Other researchers can test their own work.	Tool released

Part 2 — The GNN depth study, and the mistakes we caught

This is the most important part for judging whether the work is sound. We made errors, found them ourselves, and fixed them. Every result before step 16 was discarded.

Step	What we did	What we found	What it meant	Outcome
13	Tested deeper graphs 1, 2, 3 and 4 layers of neighbour-sharing.	First run looked clean: plain GNN got worse with depth, gated GNN stayed flat.	We nearly wrote this up. It was wrong.	Result later voided
14	Re-ran the same command	Completely different numbers. Same settings gave 48.86 one day and 54.03 the next. One setting gave 46.79 once and 0.00 the next.	The results were not reproducible. Nothing could be published in that state.	Serious problem found
15	Found the cause	Only one random seed was being set. The GPU was still free to vary between runs.	A reviewer who could not reproduce our tables would have ended the submission.	Root cause
16	Fixed it Full seeding, deterministic GPU settings, gradient clipping, guard against NaN.	Ran the same setup twice: identical to 6 decimal places. Later held across a machine restart.	From here, every number is reproducible. All earlier numbers discarded.	PASS
17	Found a second, separate mistake	The 'no graph' comparison used 1 layer while graph models used up to 4. We were comparing unequal models.	Any difference could have been caused by size, not by the graph. Fixed by pairing each depth with its own equal-sized control.	Comparison corrected
18	Re-ran everything properly 3 graph types x 4 depths x 5 seeds, both cities.	Los Angeles: 12 of 12 comparisons negative, 9 significant. Chicago: 10 of 12 negative.	The graph never helped. This is now solid and reproducible.	Clean negative result

Part 3 — The final experiment, and the main finding

Step	What we did	What we found	What it meant	Outcome
19	Spotted something odd The plain GNN at depth 2 seemed to help the poorest areas.	Tail score rose +7.97 in LA and +19.24 in Chicago. Both statistically significant.	It looked like we had solved the fairness problem that three earlier methods failed on.	Looked like a win
20	Did not trust it. Checked with AUC AUC cannot be fooled by base rates or by the decision threshold.	Tail F1 +16.95 , but Tail AUC -4.01 . True-positive rate jumped from 23% to 80%. Precision fell 6.31.	The model was not better. It had simply started predicting 'crime' almost everywhere in poor areas.	It was an illusion
21	Checked our own correction	Our base-rate correction said the gap closed from 40.97 to -2.39 — a seemingly perfect result. It was fooled too.	Correcting for base rate is not enough . A model can fake a fairness win by shifting its threshold. You must also report AUC.	Main contribution

Part 4 — The headline numbers

Chicago, plain GNN with 2 layers, compared against an equal-sized model with no graph. 5 random seeds, fully reproducible.

Measure	No graph	With graph	Change	Plain-English meaning
Tail F1 (the usual fairness measure)	23.42	40.38	+16.95	Looks like a large improvement
Tail AUC (cannot be fooled)	62.03	58.02	-4.01	Actually got worse
Tail true-positive rate	23.46	79.91	+56.45	The explanation: it now says 'crime' 80% of the time
Tail precision	34.27	27.96	-6.31	More of those predictions are wrong
Head F1	85.20	84.64	-0.56	Busy areas unchanged
Base-rate-corrected gap	40.97	-2.39	-43.36	Our own correction was fooled as well

In one sentence: a change that appears to make the model much fairer to poor neighbourhoods is actually the model labelling most of those neighbourhoods as high-crime, while genuinely getting worse at telling which ones really are. In predictive policing, that is the opposite of a fairness improvement.

Part 5 — Where things stand

Item	Status	Notes
Theory (the $2p/(1+p)$ result)	Done	Not ours. Known since Davis & Goadrich 2006, Boyd 2012, Chouldechova 2017. Must be cited as background, never claimed.
Audit of 7 published papers	Done	New and checkable. A real contribution.
Effect on decisions	Done	New. Shows the flaw has practical cost.
fairaudit tool	Done	Supports the paper; not a contribution on its own.
GNN artifact demonstration	Done	Strongest result. The flaw caught live in a working model, on two cities.
'Correction is not enough'	Done	Best novelty claim. Non-obvious and actionable.
Gated GNN as a method	Done	Did not work — negative on both cities. Cannot be the paper's contribution.
Attack / defence experiments	Void	Produced with the old faulty code. Would need re-running. Recommend leaving out — it is a separate paper.
The manuscript	Not started	The real remaining work. Roughly 4-6 weeks.
Ethics / limitations section	Not started	Required for anything involving predictive policing.

A note on the mistakes

Steps 13-17 record errors that were found and corrected before publication rather than after. Two results that looked good were withdrawn: the first depth-ablation table, and the apparent fairness improvement in step 19. Catching both internally is the reason the remaining numbers can be defended. Any result in this project produced before step 16 should be treated as void.